

KSHITIJ PRITISH BHARAMBE

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SUMMARY

Backend-focused Software Engineer with 4+ years of experience designing and shipping distributed systems, cloud-native microservices, and production-grade AI integrations at high scale. Proven track record of owning end-to-end features from architecture through deployment; reduced system latency by **40%** and drove **\$2M+** in measurable engineering impact across fintech and enterprise SaaS domains. **AWS** Certified Solutions Architect with deep expertise in **Python**, **FastAPI**, **Kafka**, and agentic AI systems built on **LangChain** and **MCP**. At **Intuit**, led backend modernization initiatives powering **QuickBooks** and **TurboTax** platform services serving **100M+** users.

WORK EXPERIENCE

Intuit

Jan 2025 – Present

Software Engineer II, QuickBooks Platform

Mountain View, CA

- Architected a real-time financial data synchronization service in **Python** using **Kafka** event streams and **Cassandra** for state persistence; reduced transaction-sync propagation latency by **38%** across 12 regional data centers serving **8M+** QuickBooks Online users
- Designed and deployed an agentic AI bookkeeping assistant integrating **LangChain** agents with internal **MCP** (Model Context Protocol) tooling; improved automated categorization accuracy by **14%** for 3 high-volume SMB cohorts within 8 weeks of rollout
- Built a GenAI-powered expense-forecasting microservice on **AWS SageMaker** with **Lambda** inference endpoints; cut cold-start latency by **52%** and reduced over-accrual events by **22%** across TurboTax Business filings
- Migrated 6 legacy **Python** monolithic services to **FastAPI**-based microservices on **AWS EKS**; eliminated 3 critical SLA breaches per quarter and reduced infrastructure spend by **\$180K/year**
- Implemented distributed tracing pipeline using **Jaeger** and **Prometheus** across 18 downstream services; cut mean time-to-detection (MTTD) for incidents by **45%** and reduced on-call pages by **30%**
- Authored internal engineering RFC adopted by 4 teams for async-safe **Kafka** consumer patterns with idempotent processing; prevented 3 duplicate-payment incidents and improved consumer throughput by **2.1x**

Razorpay

Oct 2021 – Jul 2023

Software Engineer II

Bengaluru, India

- Owned backend architecture for Razorpay's Smart Collect reconciliation service using **Python (FastAPI)** and **PostgreSQL**, with distributed systems patterns including event sourcing and idempotent retry queues; processed **4M+** daily transactions with **99.97%** uptime across 18 months
- Delivered end-to-end payment-routing ML pipeline: built **Python** feature engineering jobs on **Apache Spark**, trained gradient-boosted classifier, and served predictions via **Flask** API; improved routing success rate by **9.2%** translating to **\$1.4M** in recovered GMV annually
- Designed multi-cloud failover strategy across **AWS** and **GCP** for critical payment APIs; achieved RPO of **30s** and RTO of **90s**, eliminating 2 regional outage events in H2 2022
- Built automated fraud-signal ingestion pipeline using **AWS Kinesis** and **Redis** caching layer; cut fraud-check p99 latency from **320ms** to **48ms** and blocked **12K+** fraudulent transactions in first 60 days

Razorpay

Feb 2021 – Sep 2021

Associate Developer

Bengaluru, India

- Integrated Razorpay Payments API into 3 enterprise merchant onboarding flows using **Node.js** and **MySQL**; reduced integration time from 5 days to 18 hours per merchant
- Contributed to internal developer tooling for API contract testing using **Pact** framework; reduced regression failures in CI/CD by **41%** across 7 microservices
- Revamped frontend dashboard for merchant payout analytics using **React** and **TypeScript**; reduced merchant support ticket volume by **28%** and cut average load time from **4.1s** to **1.2s**
- Promoted to Software Engineer II within 11 months; recognized for owning 3 production microservices independently and driving zero-downtime deployments via **Kubernetes** blue-green rollouts

PROJECTS

AgentFlow: Multi-Agent Task Orchestration Framework

Python, LangChain, FastAPI, Redis, Docker

- Built a multi-agent orchestration system using **LangChain** with **MCP** tool integrations and **FastAPI** async APIs; load-tested to 50+ concurrent agent sessions with sub-**200ms** task dispatch latency via **Locust** benchmarks
- Implemented dual-layer agent memory: **Redis** vector store for low-latency in-session context retrieval and **pgvector** for durable cross-session long-term memory; improved task completion accuracy by **31%** over stateless baseline on

1,000-task benchmark suite

SmartPipe: Real-Time ML Data Pipeline

Python, Kafka, Spark Streaming, AWS S3, Airflow

- Designed an end-to-end streaming ML pipeline ingesting **50K** events/sec using **Kafka** and **Spark Streaming**; automated feature store refresh via **Apache Airflow** DAGs with zero manual intervention
- Benchmarked streaming architecture against equivalent **Airflow** batch baseline on identical dataset; reduced data-to-model feedback loop from 6 hours to 22 minutes with documented benchmark scripts in repo

DistributedKV: Fault-Tolerant Key-Value Store

Python, Raft Consensus, gRPC, Docker

- Built a distributed key-value store from scratch in **Python** implementing the **Raft** consensus algorithm with leader election, log replication, and snapshotting; maintained linearizable reads and writes across a 3-node cluster under simulated network partitions
- Designed a **gRPC**-based client interface with retries and leader-redirect logic; validated correctness using Jepsen-style chaos tests injecting random node failures and network delays across **500+** test iterations

EDUCATION

Syracuse University

Master of Science, Computer Science

Aug 2023 – May 2025

Syracuse, NY

New Horizon Institute of Technology and Management

Bachelor of Engineering, Computer Engineering

Aug 2019 – May 2023

Thane, India

SKILLS

Languages: Python, JavaScript/TypeScript, SQL, Bash

Backend & APIs: FastAPI, Flask, Django, Node.js, REST, GraphQL, Microservices Architecture

AI/ML: LangChain, MCP (Model Context Protocol), LLM Integration, AWS SageMaker, Apache Spark, Feature Engineering, Agentic AI Systems

Cloud & DevOps: AWS (EKS, Lambda, Kinesis, S3, SageMaker, RDS), GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Blue-Green Deployments

Data & Messaging: Kafka, Apache Spark Streaming, Apache Airflow, PostgreSQL, MySQL, Cassandra, Redis, pgvector, MongoDB

Observability: Prometheus, Grafana, Jaeger, Datadog, Distributed Tracing, Incident Response

Certifications: AWS Certified Solutions Architect Associate